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Pileup correction algorithm using an iterated sparse reconstruction method

Tom Trigano, *Member, IEEE,* Ilia Gildin, and Yann Sepulcre

Abstract

Every radioactive source can be characterized by an histogram obtained after collecting the energies of photons emitted from the source, also called energy spectrum. However, when the activity of this source is high, a physical phenomenon known as the pile-up effect distorts direct measurements, resulting in a significant distortion of the energy spectrum. We suggest in this paper an iterative algorithm to attenuate the pile-up effect and enhance the resulting energy spectra. It is based on iterations of a post-processed, non-negative, version of the Least Absolute Shrinkage and Selection Operator (LASSO). Results on simulations and real data illustrate the improvement obtained by the proposed method.

Index Terms

Compressive sensing, sparse reconstruction, nuclear spectroscopy, signal restoration and enhancement.

I. Introduction

A gamma spectrometer apparatus usually consists of a semiconductor detector, on which photons impinge and create pairs of electron-holes. Their migration creates pulses, characterized by a fast rise and a slow decay, whose energy is recorded by a spectrum analyzer. The histogram of the recorded energies (known in this framework as an *energy spectrum*) allows identification of unknown radioactive sources. Due to the finite resolution of the detector, close electrical pulses may overlap, causing a distortion known as "pileup effect" in the energy spectrum and the activity estimation [1]. Pileup effect may occur

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if the radioactive activity of the source is high. Figure 1 illustrates the pileup effect in the time domain. The energies of the pileups create a distortion in the energy spectra, and makes the identification of the

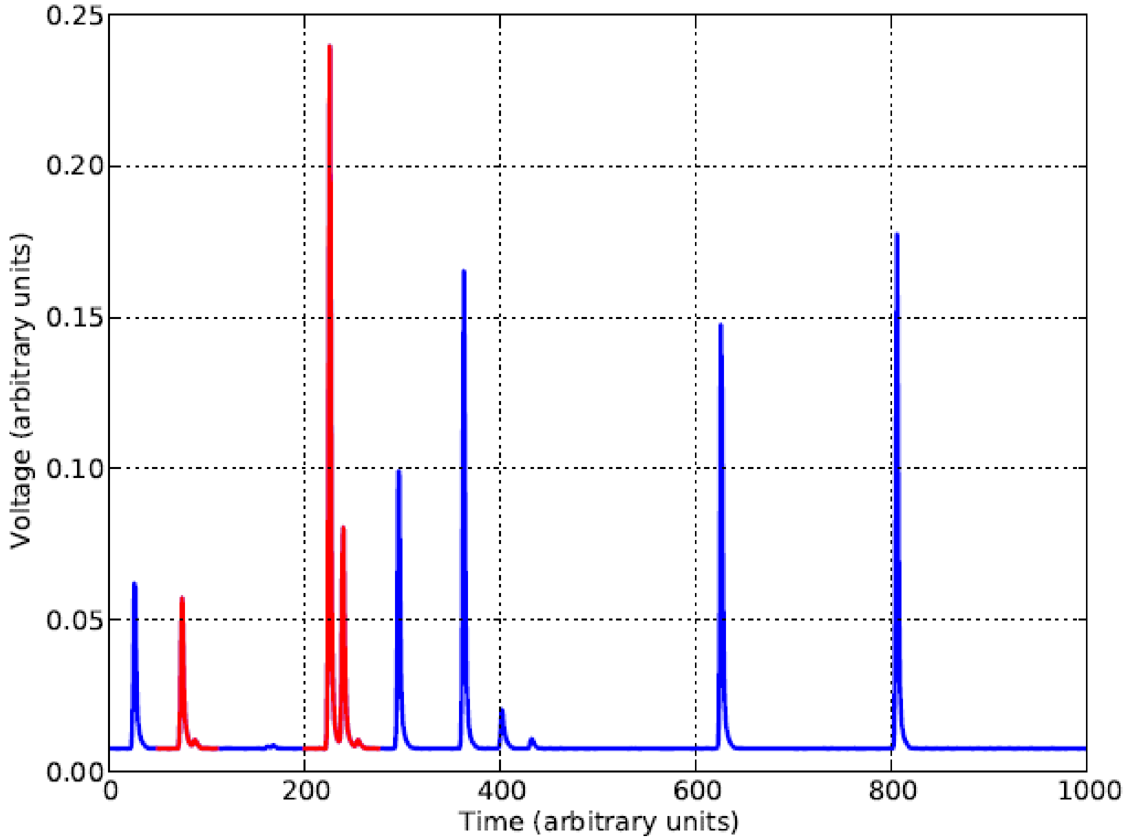

Fig. 1. Example of real signal stemming from a spectrometer. Some pileups are emphasized in red.

radioactive sources harder.

In this paper, we present a new algorithm to attenuate the pileup effect in order to obtain better energy spectra. Standard approaches for pileup correction are usually based on hardware pileup rejectors [2], and usually make strong assumptions on pulse durations in order to provide satisfactory results. Recently, [3] developed an analytical pileup correction formula to process the distorted energy spectra offline. Though efficient, its computational complexity prevents its implementation for real time apparatus. On the other hand, it can be understood from Figure 1 that the signal at hand is, in many cases, sparse, and could be processed by means of compressive sensing techniques [4]. Recently, [5] suggested a method based on the Least Absolute Shrinkage and Selection Operator (LASSO) to attenuate the pileup effect. The present paper presents an improvement of [5].

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The rest of the paper is organized as follows. Section II recalls the model used alongside the paper, and suggests an algorithm based on weighted iterations of a non-negative version of LASSO for the computations of individual energies. Results on simulations and real data are detailed in Section III.

II. Algorithm for pile-up correction

A. Model

The recorded, sampled with a sampling period of Δt, signal is modeled by

$$y[n] = \sum_{k=0}^{\infty} E_k \Phi_k[n - T_k] + \varepsilon_n, n = 0 \ldots N-1 \tag{1}$$

where $\{E_k, k \geq 0\}$ is a sequence of independent and identically distributed random variables with common density function f_E, $\{T_k, k \geq 0\}$ denotes the arrival times of the electrical pulses, and are modeled by a homogeneous Poisson process with unknown intensity λ, Φ_k denotes the shape of the k-th electrical pulse, and ε_n denotes an additive Gaussian noise with known variance σ^2. Though the shapes Φ_k are unknown, a standard recorded gamma pulse is usually characterized in practice by a fast voltage increase due to the creation of the electron-hole pairs, followed by a slower exponential decay caused by the migration of these pairs over a finite duration. A decent assumption is therefore to model Φ_k by a truncated gamma shape:

$$\Phi_{(\theta_1^{(k)}, \theta_2^{(k)})}[n] = n^{\theta_1^{(k)}} e^{-\theta_2^{(k)} n} 1(0 \leq n \leq N-1), \tag{2}$$

where $\theta_1^{(k)}$ and $\theta_2^{(k)}$ are parameters controlling the pulse shape. Their range is usually known (accordingly to the detector's dynamics) but not their exact value. Further on we shall write that (θ_1, θ_2) belongs to $(0, \Theta_1) \times (0, \Theta_2)$, and shall assume that the values of $\Theta_1,\ \Theta_2$ are known. The problem to address is the estimation of f_E, given the observations $y[n], n = 0 \ldots N-1$.

We choose in this paper not to assume any parametric model regarding f_E, and therefore to estimate it in a non-parametric setting. Since the E_k's are unknown in practice, they must be estimated first from $y[n]$. From the specific shape of (1), it can be seen that the estimation of the E_k's is strongly related to a sparse regression problem, as presented below.

Denote by $\{s_1, \ldots, s_{m_1}\}$ (respectively $\{p_1, \ldots, p_{m_2}\}$) a uniform sampling of $(0, \Theta_1)$ (resp. $(0, \Theta_2)$), and by $\mathbf{A}_n$ the following collection of potential pulse shapes for any $n = 0 \ldots N-1$:

$$\mathbf{A}_n = [\Phi_{(s_{k_1}, p_{k_2})}[k-n]]^T,$$
$$1 \leq k_1 \leq m_1,\ 1 \leq k_2 \leq m_2,\ 0 \leq k \leq N-1. \tag{3}$$

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Denote by $\mathbf{A}$ the dictionary obtained by concatenation of the $\mathbf{A}_n$ from (3):

$$\mathbf{A} = [\mathbf{A}_0 \ \mathbf{A}_1 \ \ldots \ \mathbf{A}_{N-1}]. \tag{4}$$

Thus, (1) can be rewritten as

$$\mathbf{y} = \mathbf{A}\boldsymbol{\beta} + \boldsymbol{\varepsilon} + \boldsymbol{\delta}, \tag{5}$$

where we define $\mathbf{y} \triangleq [y[0] \ y[1] \ \ldots \ y[N-1]]^T$, $\boldsymbol{\varepsilon} \triangleq [\varepsilon[0] \ \varepsilon[1] \ \ldots \ \varepsilon[N-1]]^T$, $\boldsymbol{\delta}$ is the approximation error obtained by replacing the T_k's by the closest sampling points and (1) by a discrete linear model as in (5) (we refer to [6] for an in-depth discussion on $\boldsymbol{\delta}$) and $\boldsymbol{\beta}$ is a regressor containing the information regarding the E_k's. Due to the sparse nature of the signal $\mathbf{y}$, the coefficients of $\boldsymbol{\beta}$ are mostly equal to zero, and are non-zero around arrival times only. Therefore, it is possible to relate the estimation of the E_k's to the sparse estimation of $\boldsymbol{\beta}$ in (5). Further on, we will decompose $\boldsymbol{\beta}$ blockwise, accordingly to (4), and write $\boldsymbol{\beta} \triangleq [\boldsymbol{\beta}_0^T \ \boldsymbol{\beta}_1^T \ \ldots \ \boldsymbol{\beta}_{N-1}^T]^T$; we will also denote by $P \triangleq m_1 m_2$ the overall number of different shapes in a given block $\mathbf{A}_k$, so that $\mathbf{A}$ is of size $N \times NP$.

B. Overview of the Algorithm

LASSO (Least Absolute Shrinkage and Selection Operator) was introduced in [7], to overcome the fact that the ordinary least square estimate fails to identify a sparse regressor, and its non-negative version is defined as

$$\hat{\boldsymbol{\beta}}(r) \triangleq \underset{\boldsymbol{\beta} \geq 0}{\arg\min} \left\{ \|\mathbf{y} - \mathbf{A}\boldsymbol{\beta}\|_2^2 + r \|\boldsymbol{\beta}\|_{\ell_1} \right\}, \tag{6}$$

where the tuning parameter r quantifies the trade off between sparsity and estimation precision. (6) is solved efficiently by the LARS algorithm [8], which makes it well fitted for real-time applications. However, it performs poorly in variable selection tasks when there are high correlations between active and inactive variables. Several papers (e.g. [9], [10]) underline that very restrictive conditions must hold to get true selection consistency, and these conditions are not verified in our framework. However, it remains possible to estimate precisely the arrival times of each individual electrical pulse by means of a post-processed non-negative LASSO [6].

The algorithm proposed in this paper can be decomposed as follows: we first estimate the arrival times of the signal by means of the method described in [6]. In a second step, a reweighted form of the non-negative LASSO is applied iteratively on the signal in order to find the best approximation for each pulse separately. The motivation for an iterative approach is that LASSO tends to optimize the global error on the whole signal, and does not focus on specific parts. It is therefore legitimate to think that a

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sparsity parameter suitable for one given pulse will not be fit for another one, thus each pulse should be investigated separately. Another reason lies in the fact that global optimization yields underestimated energies, since some region overlap. It is important to follow the whole chronology of the incoming pulses in order to estimate these energies as precisely as possible.

C. Estimation of the Arrival Times

The estimation of the arrival times is performed as in [6], whose procedure is now briefly recalled: denote by $\hat{\boldsymbol{\beta}}(r)$ the regressor obtained by solving (6) (the sparsity parameter r is chosen as in [6]). We further on decompose $\hat{\boldsymbol{\beta}}(r)$ blockwise accordingly to (4), that is

$$\hat{\boldsymbol{\beta}}(r) = \left[\hat{\boldsymbol{\beta}}_0^T(r)\ \hat{\boldsymbol{\beta}}_1^T(r) \dots\ \hat{\boldsymbol{\beta}}_{N-1}^T(r)\right]^T .$$

Provided $\lambda \Delta t \ll 1$, which is often the case in practice, consecutive blocks $\hat{\boldsymbol{\beta}}_k(r)$ with a sufficiently large ℓ_1-norm are most likely related to the same arrival time. Therefore, the arrival times are estimated recursively by:

$$\begin{cases} \widehat{T}_0 & = \min\Big\{0 \le m \le N-1;\ \|\hat{\boldsymbol{\beta}}_m(r)\|_1 \ge \eta\Big\}, \\ \widehat{T}_k & = \min\Big\{m > \widehat{T}_{k-1}; \|\hat{\boldsymbol{\beta}}_m(r)\|_1 \ge \eta, \\ & \quad \|\hat{\boldsymbol{\beta}}_{m-1}(r)\|_1 < \eta\Big\}, \end{cases} \tag{7}$$

where η is a user-defined threshold which guarantees that only the most representative blocks are used for the estimation of arrival times. Equation (7) is motivated by the fact that the columns of $\mathbf{A}$ are highly correlated, and therefore consistency in selection and irrepresentability condition in the sense of [10] cannot be attained. Thus, (7) controls the fact that clusters of consecutive selected blocks are related to a single event, and discards the less significant blocks of $\hat{\boldsymbol{\beta}}(r)$ in the ℓ_1 sense, since these blocks contribute more to refining the estimation of the pulses' shapes rather than their arrival times.

D. Energies and Energy Spectrum Estimation

After estimating the arrival times of the pulses, we now estimate the shape of the pulses individually. This can be achieved by multiplying the dictionary by a weighting diagonal matrix $\mathbf{W}$ which will act as a sieve around the estimated arrival times. In order to respect the chronology of the arrival times, we shall, at each iteration, apply the non-negative LASSO on a residual, that is the difference between the signal and the previously estimated electrical pulse.

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Mathematically, the algorithm is summarized as follows: denote by $\mathbf{y}^{(k)}$ the k-th residual, that is, the remaining signal after we already estimated $k-1$ energies of the $k-1$ previous electrical pulses associated with $\widehat{T}_0, \dots, \widehat{T}_{k-1}$ (we set $\mathbf{y}^{(0)} \triangleq \mathbf{y}$). We define the following associated weighted diagonal matrix:

$$\mathbf{W}^{(k)} = \mathrm{diag}(w_i^{(k)}),$$

$$w_i^{(k)} \triangleq c_k \exp\left\{ -\frac{(i-\mu_k)^2}{2\sigma_k^2} \right\}, i = 1 \dots NP \quad (8)$$

where $\mu_k \triangleq P\widehat{T}_k + \arg\max \hat{\boldsymbol{\beta}}_{\widehat{T}_k}(r)$ is used to localize the most significant weights around the coefficients of $\hat{\boldsymbol{\beta}}(r)$ associated with $\widehat{T}_k$, c_k is a normalizing constant chosen so that $\mathrm{Tr}(\mathbf{W}^{(k)}) = 1$, and the parameter σ_k depends on the distance between consecutive arrival times, in order not to include columns of $\mathbf{A}$ used in other pulses. In other words, we use at the k-th iteration the dictionary $\mathbf{A}\mathbf{W}^{(k)}$ in order to use the non-negative LASSO locally near the k-th arrival time. The energy of the electrical pulse associated with $\widehat{T}_k$ is then estimated in two steps:

1) We first compute a non-negative LASSO for the electrical pulse associated with $\widehat{T}_k$:

$$\hat{\boldsymbol{\beta}}(r_k) = \arg\min_{\boldsymbol{\beta} \geq 0} \left\{ \|\mathbf{y}^{(k)} - \mathbf{A}\mathbf{W}^{(k)}\boldsymbol{\beta}\|_2^2 + r_k \|\boldsymbol{\beta}\|_1 \right\}; \quad (9)$$

2) we then estimate its energy as

$$\hat{E}_k = \|\mathbf{A}\mathbf{W}^{(k)}\hat{\boldsymbol{\beta}}(r_k)\|_1$$

The next residual is then computed as

$$\mathbf{y}^{(k+1)} = \mathbf{y}^{(k)} - \mathbf{A}\mathbf{W}^{(k)}\hat{\boldsymbol{\beta}}(r_k),$$

and the procedure is repeated until all arrival times have been investigated. Eventually, the corrected energy spectrum is estimated by plugging the values $\hat{E}_k$ in a standard kernel estimate:

$$\hat{f}_E(x) \triangleq \frac{1}{Mh} \sum_{k=1}^{M} K\left(\frac{x - \hat{E}_k}{h} \right), \quad (10)$$

where M denotes the number of found arrival times, K is a kernel satisfying standard conditions (e.g. integrability to one), and h is a bandwidth parameter controlling the smoothness of the estimate.

III. Results

In this section, we present results of simulations as well as on real data. The presented algorithm depends on several user-defined parameters which must be carefully calibrated in order to provide

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acceptable results. We refer to [6] for the in-depth discussion on the choice of the sparsity parameter r and of the threshold η for the estimation of the arrival times, and choose $\eta = 3\sigma$ and r so that $\|\mathbf{y} - \mathbf{A}\hat{\boldsymbol{\beta}}(r)\|_2 \leq 3\sigma\sqrt{N}$. When estimating the energy of the k-th electrical pulse, the variance σ_k of the weighting matrix $\mathbf{W}^{(k)}$ must be chosen accordingly to the close arrival times, since the reweighted dictionary $\mathbf{AW}^{(k)}$ must not include columns related to the $k-1$-th or the $k+1$-th electrical pulse. Therefore we chose in our applications

$$\sigma_k = 0.5 \min \left\{ \widehat{T}_k - \widehat{T}_{k-1}, \widehat{T}_{k+1} - \widehat{T}_k \right\} , \tag{11}$$

and $\sigma_0 = 0.5 \min \left\{ \widehat{T}_0, \widehat{T}_1 - \widehat{T}_0 \right\}$, where the coefficient 0.5 is introduced in order to ensure that the tails of the Gaussian shape will be numerically zero around the blocks of $\hat{\boldsymbol{\beta}}(r)$ associated with $\widehat{T}_{k-1}$ and $\widehat{T}_{k+1}$.

The sparsity parameter r_k in (9) must guarantee sufficient precision in the ℓ_2 sense, for a good estimation of the energies. We suggest in this paper to take r_k so that $\|\mathbf{y}^{(k)} - \mathbf{AW}^{(k)}\hat{\boldsymbol{\beta}}(r_k)\|_2 \leq 6\sigma\sigma_k$., though other choices would be legitimate for other detectors with different Signal Noise Ratios (SNRs). The choice of the kernel function in (10), as stated in [11] is not a critical issue, and is taken Gaussian in our application. Though numerous automatic bandwidth selectors exist in the literature [12], its choice is in practice driven by physical considerations, such as the resolution of the spectrometric apparatus. In this paper we chose $h = 5$ for the simulations, and $h = 1$ for the real data investigation.

A. Simulations

As previously discussed, our algorithm will use a dictionary $\mathbf{A}$ in order to estimate the real signal. Thus, in our simulations, we create the artificial signal directly out of the dictionary. The energies are drawn accordingly to a mixture of two Gaussian distributions with means of 100 and 225 (so that pile-ups can appear in many places of the energy spectra) and standard deviations of 15 and 25. The additive noise is Gaussian with a variance equal to 0.04, in order to simulate the excellent SNR of a HPGe detector, and the shapes are drawn randomly from $\mathbf{A}$. The arrival times are drawn accordingly to a homogeneous Poisson process with intensity λ, and we investigated two different cases: $\lambda = 0.04$ and $\lambda = 0.1$, which correspond respectively to a mid-level and a very high radioactive activity. The energy spectra obtained for the three cases described are presented in Figure 2.

B. Results on Real Data

The performances of the proposed approach were also investigated on real data stemming from a gamma spectrometer apparatus. We performed experiments on a mixture of Cesium 137 and Americium

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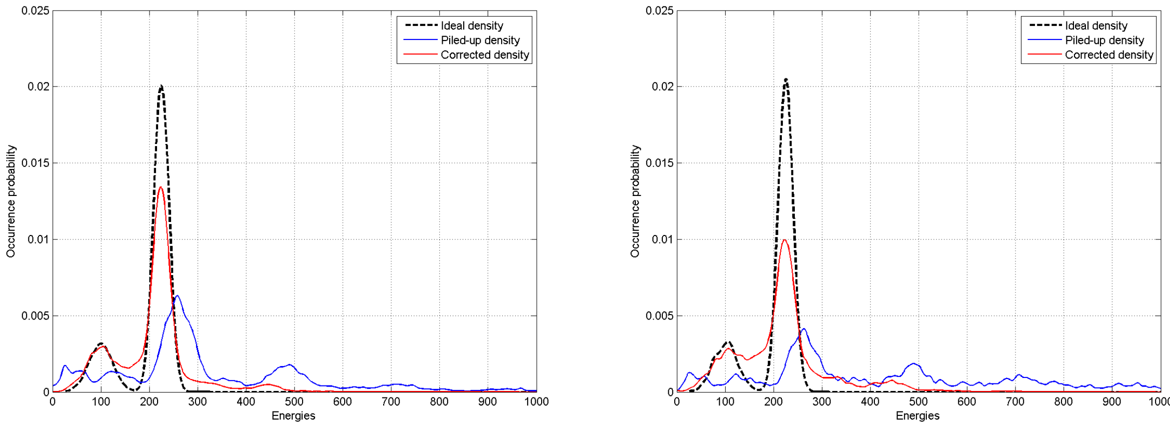

(a) $\lambda = 0.04$ (b) $\lambda = 0.1$

Fig. 2. Results on simulations

241. The first radioactive element is characterized by a single peak in its spectrum at 667 keV, and is used for calibration. Americium 241 energy spectrum includes a large number of peaks in its energy spectrum, thus generating many pileups. Given that the location of Americium peaks at low activity is cumbersome (they are hardly separable from the Compton continuum of Cesium 137), we chose to illustrate the performances of the algorithm by comparing it to a standard thresholding method (that is, an energy is recorded when the signal goes above a user-specified threshold, which means that no pileup correction is performed), which is commonly used in the field. The results obtained on real data are given in Figure 3. It can be observed that the double peak of Cesium and a significant part of the replicated Compton continuum have been corrected. Nevertheless, some higher energies have not been corrected.

C. Discussion

It may be observed, in both simulations and real data, that around real peaks the energy distribution seems overestimated, and that high energy locations of the spectra are less corrected than other areas associated with lower energies. That can be easily interpreted because, as the activity of the source becomes higher, several pulses are likely to start between two consecutive sampling points. Thus, the arrival times estimation step may be misled in treating both as one single impulse, which explains why some pileups are uncorrected. Though disappointing, this cannot be avoided, since it is not possible from an incomplete and highly correlated dictionary to estimate actual arrival times from a sampled version of the recorded signal. Nevertheless, the pileup distortion is well attenuated both on simulations and real

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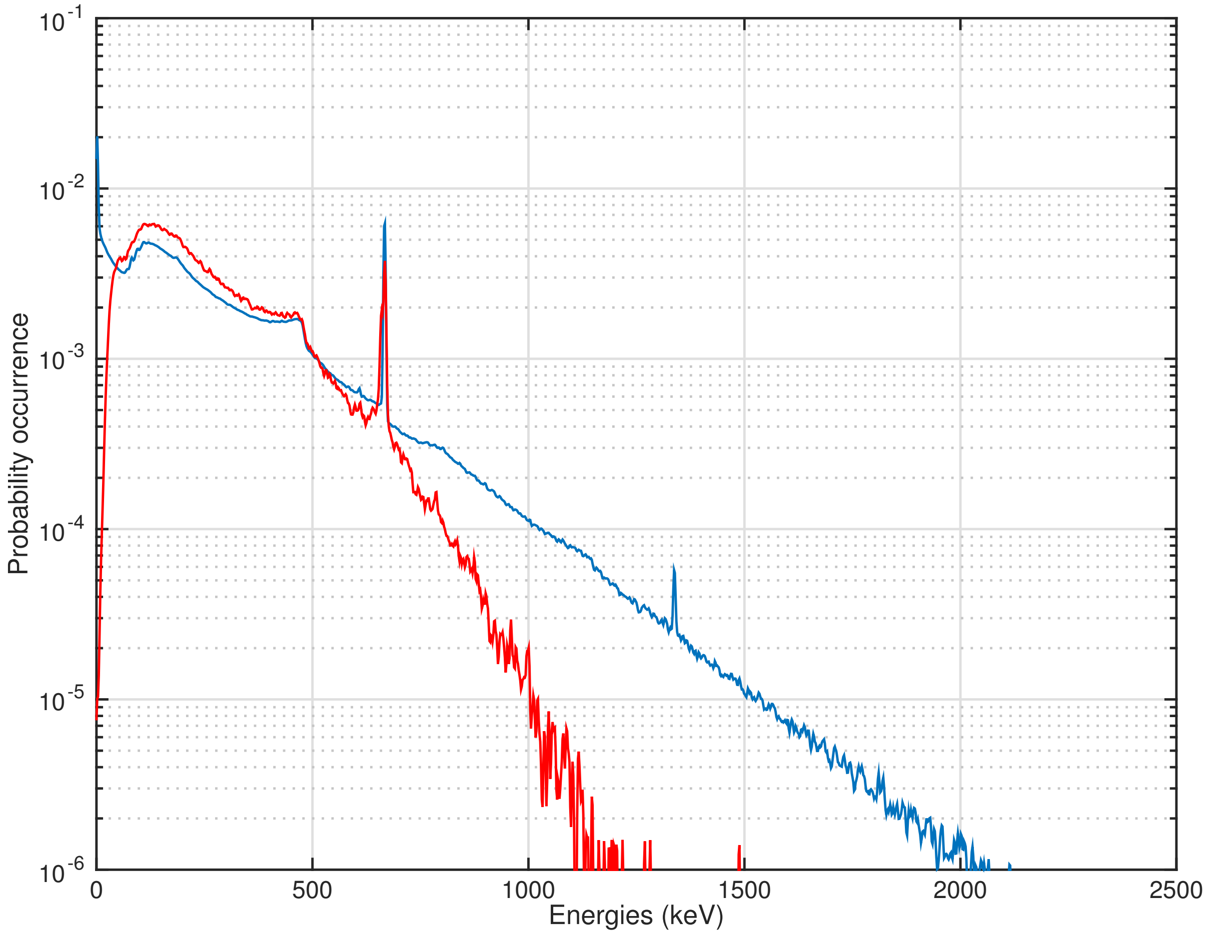

Fig. 3. Real energy spectra uncorrected (blue) and after pile-up correction (red).

data, meaning that the suggested approach can be used as a reliable pile-up attenuation technique. It can be seen from these figures that the proposed approach localizes well the real peaks of the histogram, and corrects most of the pile-up distortion in the histogram. The algorithm is surprisingly efficient even at very high activities, where most of the recorded signal consists of piled up electrical pulses. Another interesting point to notice is that the $\mathbf{W}^{(k)}$ appearing in each iteration in order to focus on the k-th pulse could be used as a pile-up rejector as well. Indeed, if (11) is small, this indicates that piled-up pulses are very close one to the other. This means that the statistical error involved in the energy estimation of the individual pulses would remain high in any case, so we should reject this pile-up and assume that the information it contains is too "corrupted" to be retrieved.

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IV. Conclusion

In this paper, we presented an algorithm for pileup attenuation for gamma spectrometric signals. The performances of this method have shown to be very promising in simulations, even with activities considered as very high. On real data, though the pileup distortion is not fully corrected, it enhances the estimation of the energy spectrum. Further investigations will detail the theoretical justifications of such an algorithm and provide quantitative results on both simulations and real data. We shall furthermore investigate the aspects related to pile-up rejection and the use of faster algorithms such as [13].

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